

QUESTION 1

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: EASY

PROMPT: If $2x + 5 = 17$, what is the value of x ?

A: 4

B: 6

C: 8

D: 11

ANSWER: B

EXPLANATION: Subtract 5 from both sides to get $2x = 12$, then divide by 2 to get $x = 6$.

END QUESTION

QUESTION 2

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: EASY

PROMPT: If $3(x - 2) = 15$, what is the value of x ?

A: 3

B: 5

C: 7

D: 9

ANSWER: C

EXPLANATION: Divide both sides by 3 to get $x - 2 = 5$, then add 2 to get $x = 7$.

END QUESTION

QUESTION 3

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: EASY

PROMPT: The equation $y = 4x + 1$ defines y in terms of x . What is the value of y when $x = 3$?

A: 9

B: 10

C: 12

D: 13

ANSWER: D

EXPLANATION: Substitute 3 for x : $y = 4(3) + 1 = 13$.

END QUESTION

QUESTION 4

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: MEDIUM

PROMPT: The system $x + y = 10$ and $x - y = 2$ has solution (x, y) . What is the value of x ?

A: 4

B: 5

C: 6

D: 8

ANSWER: C

EXPLANATION: Add the two equations to get $2x = 12$, so $x = 6$.

END QUESTION

QUESTION 5

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: MEDIUM

PROMPT: Which inequality is equivalent to $5x - 10 > 15$?

A: $x > 1$

B: $x > 5$

C: $x < 5$

D: $x \geq 5$

ANSWER: B

EXPLANATION: Add 10 to both sides to get $5x > 25$, then divide by 5 to get $x > 5$.

END QUESTION

QUESTION 6

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: MEDIUM

PROMPT: What is the slope of the line passing through the points $(2, 3)$ and $(6, 11)$?

A: 2

B: 4

C: $1/2$

D: 8

ANSWER: A

EXPLANATION: The slope is $(11 - 3) / (6 - 2) = 8 / 4 = 2$.

END QUESTION

QUESTION 7

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: MEDIUM

PROMPT: For the function $f(x) = 2x^2 - 3$, what is the value of $f(3)$?

A: 9

B: 12

C: 15

D: 18

ANSWER: C

EXPLANATION: Substitute 3 for x : $f(3) = 2(3^2) - 3 = 18 - 3 = 15$.

END QUESTION

QUESTION 8

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: HARD

PROMPT: If $4x + 3 = 2x + 15$, what is the value of x ?

A: 5

B: 6

C: 7

D: 9

ANSWER: B

EXPLANATION: Subtract $2x$ and 3 from both sides to get $2x = 12$, so $x = 6$.

END QUESTION

QUESTION 9

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: HARD

PROMPT: If $x/3 + 4 = 9$, what is the value of x ?

A: 9

B: 12

C: 13

D: 15

ANSWER: D

EXPLANATION: Subtract 4 to get $x/3 = 5$, then multiply by 3 to get $x = 15$.

END QUESTION

QUESTION 10

SECTION: MATH

CATEGORY: Algebra

DIFFICULTY: HARD

PROMPT: In the equation $y = -3x + 8$, what is the y-intercept of the line?

A: -3

B: 3

C: 8

D: -8

ANSWER: C

EXPLANATION: In slope-intercept form $y = mx + b$, the y-intercept is b.

Therefore, the y-intercept is 8.

END QUESTION